

Nicholas Pianetto

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EDUCATION

Iowa State University, College of Engineering

Expected December 2026

Bachelor of Science, Computer Engineering

SKILLS

- **Programming Languages:** C/C++, Java, Python, Verilog, VHDL
- **Embedded/Hardware:** Arduino, STM32, ESP32, Xilinx Zynq-7000, Oscilloscope, Spectrum Analyzer, Soldering (THT and SMT assembly/rework), Multimeter, Variable Power Supply, LCR Meter, Logic Analyzer, CAN/J1939, SPI, I²C, UART
- **EDA/Software:** QuestaSim, Quartus Prime, Git, PADS Logic/Layout, KiCad, STM32CubeIDE, Perforce, Lattice Diamond, FTPro, SolidWorks, Linux, Zuken, Altium Designer, Vivado, Vitis, LTspice, PetaLinux

EXPERIENCE

Electrical Engineer Intern - Vermeer Corporation

May 2026 - August 2026

- Designed and shipped a cable harness adapter to maintain pin compatibility for vendor part revision changes
- Architected an automated wire harness tester, integrating CAN bus controllers and a Linux-based display module for verification of machine harness pinouts, further reducing time for setup and testing by nearly 50%
- Managed vendor relations and component procurement, sourcing custom wire harness designs for different machines
- Conducted field testing on a machine to measure and log inrush current for cooling systems and mapped data against interconnect thermal limits to develop a mitigation strategy for overcurrent failures
- Designed the schematic in Altium Designer for a Renesas RA4M1-based system and generated a BOM for cost estimation
- Designed a PCB for a wireless ESP32-based CAN bus device to modify engine RPM during off-machine system testing, eliminating the need for a tethered CAN-to-USB interface

Circuits II Teaching Assistant - Iowa State University

January 2026 - May 2026

- Mentored students in lab with hands-on debugging, instrumentation guidance, and circuit synthesis
- Supported grading homework assignments and quizzes involving analog circuitry

Research Assistant - Iowa State University

May 2025 - May 2026

- Independently designed the schematic and PCB layout, validated functionality, and assembled an ESP32-based bioimpedance acquisition device utilizing the on-chip DFT of the AD5933 high-precision impedance converter for animal research
- Conducted meetings with stakeholders from the Department of Animal Science to establish design requirements such as built-in SPI SD card data logging and a portable LCD interface and wrote the firmware required to drive both features

Embedded Design Engineer Intern - Ag Leader Technology

May 2024 - December 2024

- Independently developed the schematic and PCB layout for a passive USB Type-C PCB with impedance matching on all differential pairs to support DisplayPort functionality for product testing during assembly
- Revised a serial to fiber optic PCB in PADS Layout and PADS Logic for use during ESD, EMC, and EMI testing
- Reworked prototype PCBs by soldering experimental shielding and 0402-sized bypass capacitors in an effort to minimize capacitive coupling problems during ESD testing and successfully mitigated identified issues

PROJECTS

PONG on Custom FPGA Development Board with VGA

- Independently designed the schematic and PCB layout, fabricated, validated, and programmed a custom FPGA board based around Lattice Semiconductor's MachXO2 7000HC FPGA, featuring onboard power regulation, configurable I/O voltage banks, synchronous 640x480 VGA timing logic, a custom FTDI FT232H-based JTAG programmer, and expansion headers; implemented a two-player PONG game in VHDL (see <https://pianetto.net/projects.html> for more details)

CPRE 488 Open-Ended Design Project (MIDI Music On Hardware)

- Developed a custom memory-mapped state-machine IP block for PWM motor control and 4-phase motor control in VHDL, and played MIDI music on floppy drives using a Xilinx Zedboard running Linux
- Reverse-engineered a flatbed scanner by analyzing datasheets of ICs on the PCB and modifying the existing PCB to actuate the 4-phase motor directly from the Zedboard's PMODs using the onboard Darlington transistor arrays and logic level shifters
- Rewrote and optimized C++ drivers to remove dependencies on large libraries, allowing for the FSBL with an I²C/SPI-based MIPI camera to fit within the 250 KB of OCM